	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2025 – 2026*

Presentation on Capstone Project Execution (CPE–316004)

Topic	Outpatient Department (OPD) Administration Management System	
Presented by	1. Sahil Mahendra Sonar	
	2. Shantanu Gokul Sonawane	
	3. Amit Ravindra Thakare	
	4. Shreya Arun Varade	
Guided by	Ms. Chaitali S. Patil	
Semester	**Sixth (K Scheme)**	**Date : 17/04/2026**

Abstract

The OPD Administration Management System is a comprehensive digital solution designed to automate the full spectrum of outpatient department operations through role-based access for administrators, doctors, and receptionists. The system facilitates patient registration and digital record maintenance, streamlines appointment scheduling, manages patient admission and discharge processes. It further extends to assign laboratory test for patients and records tracking, alongside the management of doctor and clinical staff. By digitizing the storage and retrieval of all hospital data, the platform aims to minimize manual errors, reduce wait times and significantly enhance overall administrative efficiency in a clinical setting.

Content

1. Introduction (background of the industry or used based problem / task)
2. Literature Survey
3. Scope of the project
4. Methodology
5. Details of designs, working and processes
6. Results and applications
7. Conclusion and future scope
8. References and Bibliography

1. Introduction

The Outpatient Department (OPD) is the first place where patients go when they visit a hospital or clinic. It handles many tasks like patient registration, booking appointments, doctor consultations, staff management and keeping patients records, etc. Many Clinics still use paper and pen to manage all this work. This creates several problems:

- Long waiting time for patients because registration takes too long
- Appointment confusion due to poor tracking
- Lost or misplaced patient records making it hard for doctors to check find patient
- Errors in data because of manual entry
- Poor communication between reception, doctor, and patients

An OPD Administration Management System is a software solution that solves these problems. By using a digital system, hospitals can work faster, reduce mistakes, and provide better service to patients.

2. Literature Review

A. Field Visit – Omkar Clinic, Dhule

- A field visit was conducted at Omkar Hospital, Dhule to understand the real-world workflow of OPD services
- The hospital is managed by Dr. Pranjal Patil, a cardiology specialist
- Dr. Patil holds M.D. and D.M. (Cardiology) from Mumbai
- He has more than 10 years of experience in the medical field

B. OPD Processes Observed

During the visit, the following OPD administrative processes were observed:

- Patient registration
- Appointment scheduling
- Lab patients management
- Record maintenance
- Admit patient records

2. Literature Review

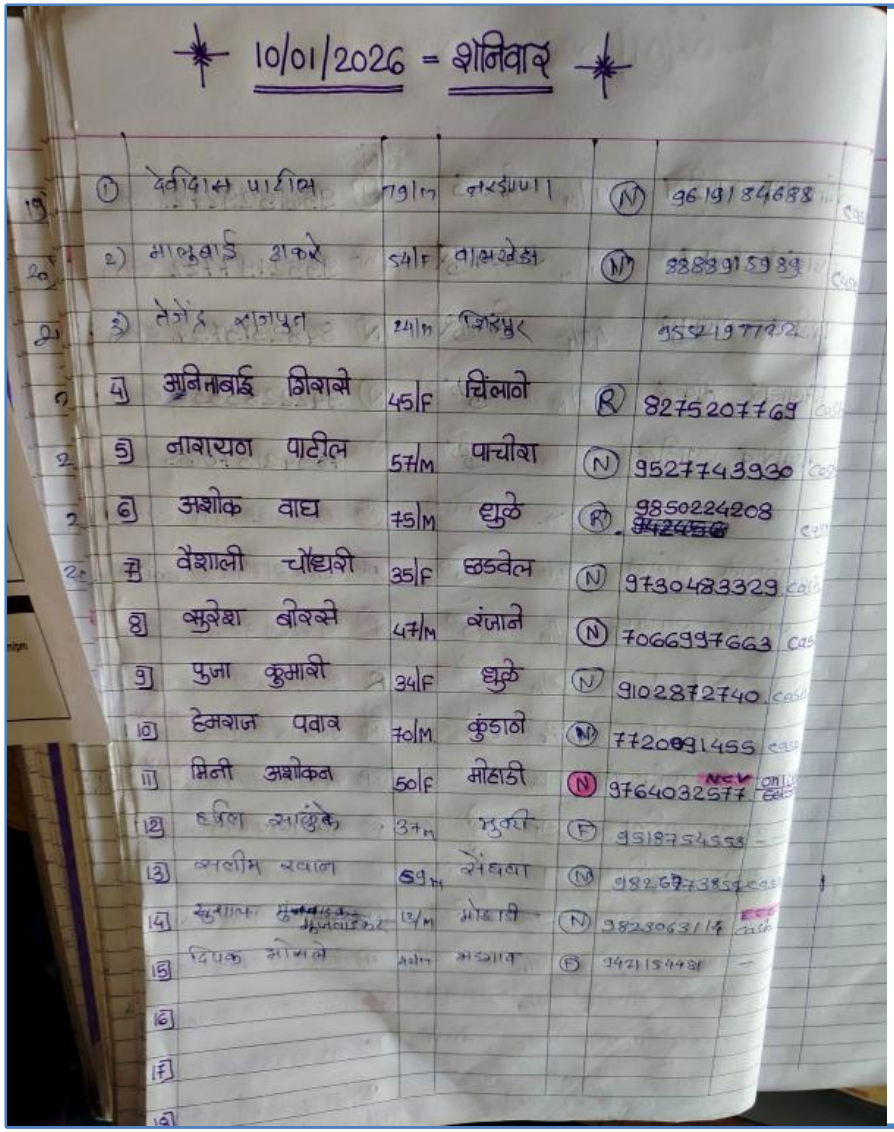
C. Challenges Identified in Manual OPD Management

- Time-consuming patient registration process
- Difficulty in tracking appointments
- Patients has to wait outside until their appointment called
- Paperwork leads to errors and data loss
- Poor coordination between staff and doctors
- Difficulty to storing data from all laboratory tests.

D. Need for Digital System

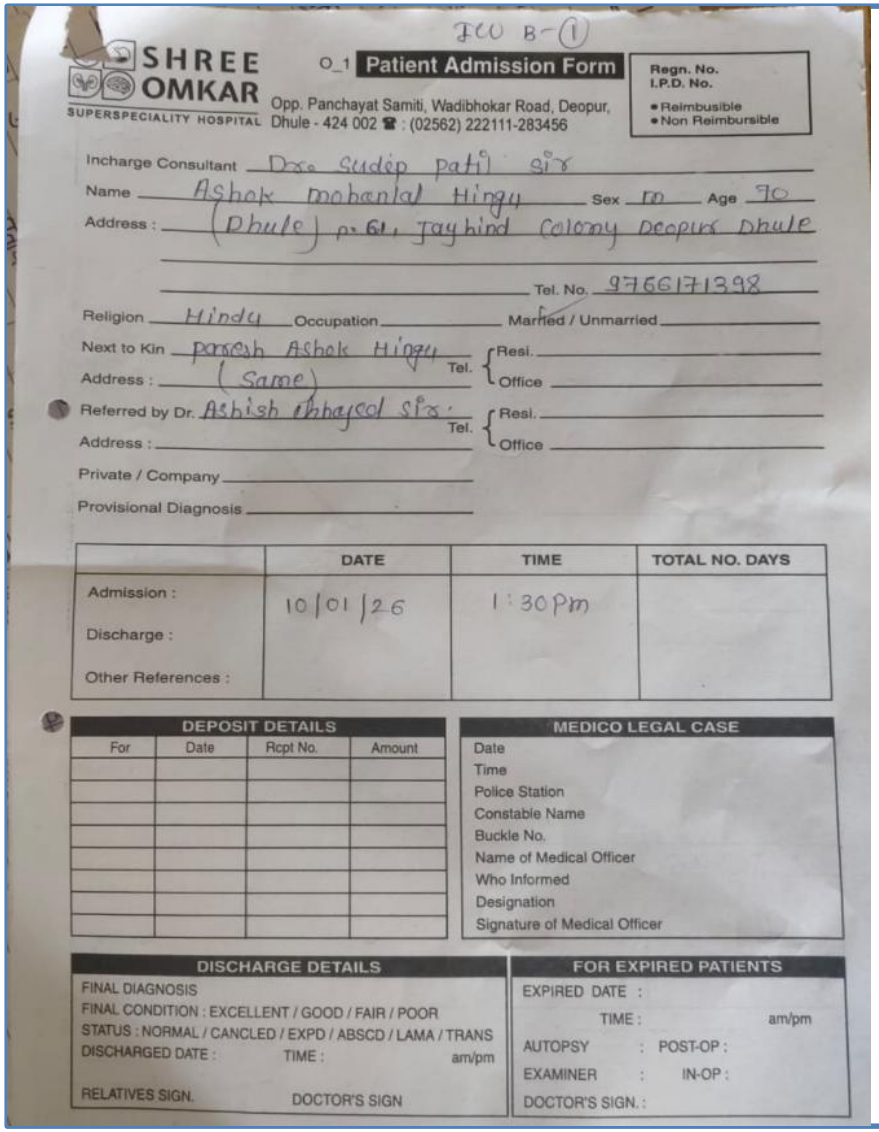
- A digital system can improve speed and accuracy
- Electronic records are easier to store and retrieve
- Better communication between all users
- Overall improvement in patient care and satisfaction

2. Literature Review



The image shows a manual appointment record captured during the field visit to Omkar Hospital. It contains handwritten entries of patient appointments, including details such as patient name, age, gender, and contact number. This manual method highlights the inefficiencies of paper-based systems, where data is difficult to organize, search, and maintain securely.

2. Literature Review



ICU B-(1)

SHREE OMKAR
SUPERSPECIALITY HOSPITAL

O_1 **Patient Admission Form**

Opp. Panchayat Samiti, Wadibhokar Road, Deopur, Dhule - 424 002 ☎ : (02562) 222111-283456

Regn. No.
I.P.D. No.
• Reimbusible
• Non Reimbursible

Incharge Consultant: Dr. Sudip patil sir
Name: Ashok mohanlal Hingu Sex: M Age: 70
Address: (Dhule) p. 61, Jayhind Colony Deopur Dhule
Tel. No. 9766171398
Religion: Hindu Occupation: Married / Unmarried
Next to Kin: paresh Ashok Hingu Tel. Resi. / Office
Address: (Same)
Referred by Dr. Ashish bhayed sir. Tel. Resi. / Office
Address:
Private / Company
Provisional Diagnosis

	DATE	TIME	TOTAL NO. DAYS
Admission :	10/01/26	1:30 Pm	
Discharge :			
Other References :			

DEPOSIT DETAILS

For	Date	Rcpt No.	Amount

MEDICO LEGAL CASE

Date
Time
Police Station
Constable Name
Buckle No.
Name of Medical Officer
Who Informed
Designation
Signature of Medical Officer

DISCHARGE DETAILS

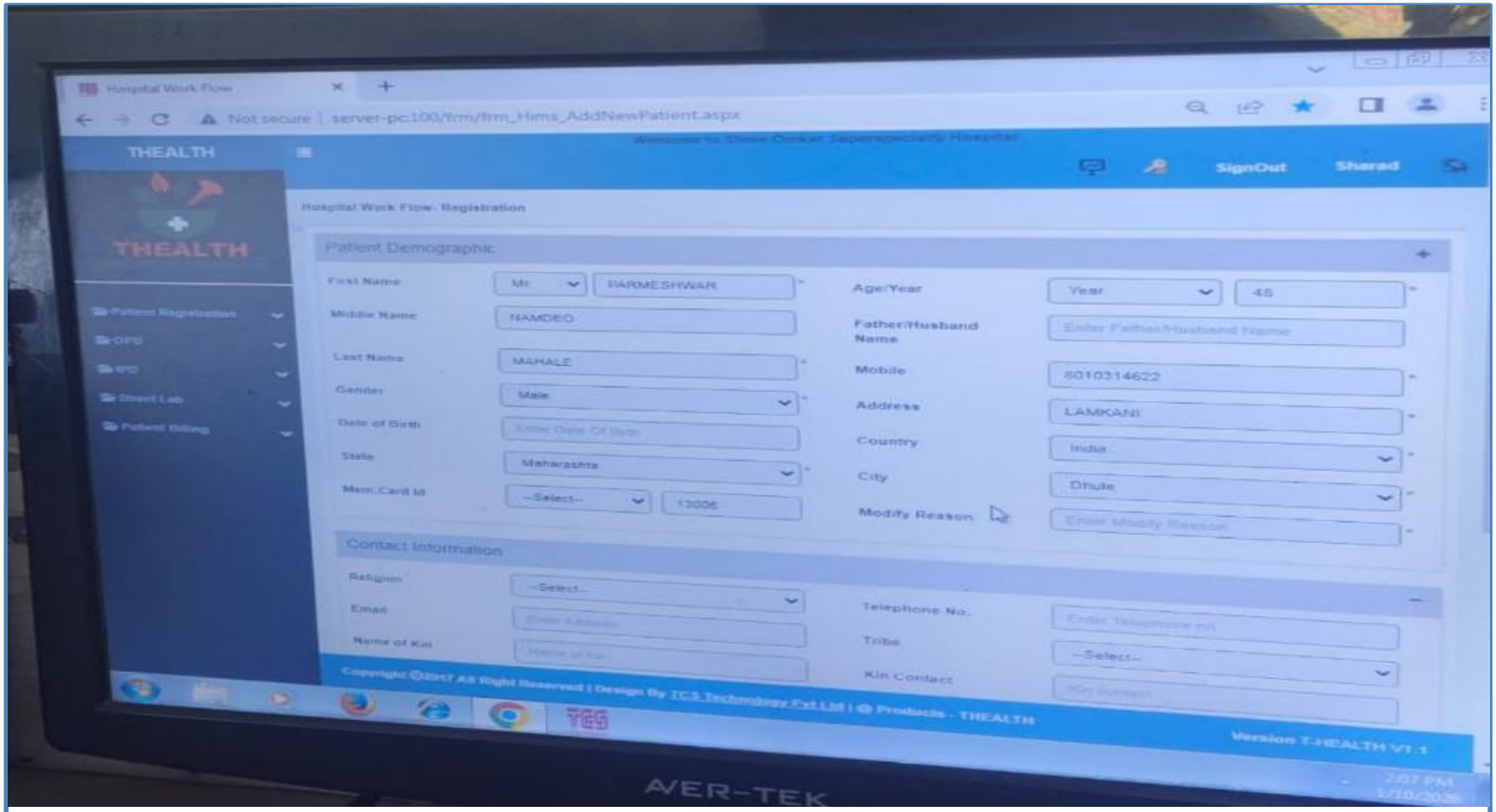
FINAL DIAGNOSIS
FINAL CONDITION : EXCELLENT / GOOD / FAIR / POOR
STATUS : NORMAL / CANCLED / EXPD / ABSCD / LAMA / TRANS
DISCHARGED DATE : TIME : am/pm
RELATIVES SIGN. DOCTOR'S SIGN

FOR EXPIRED PATIENTS

EXPIRED DATE :
TIME : am/pm
AUTOPSY : POST-OP :
EXAMINER : IN-OP :
DOCTOR'S SIGN. :

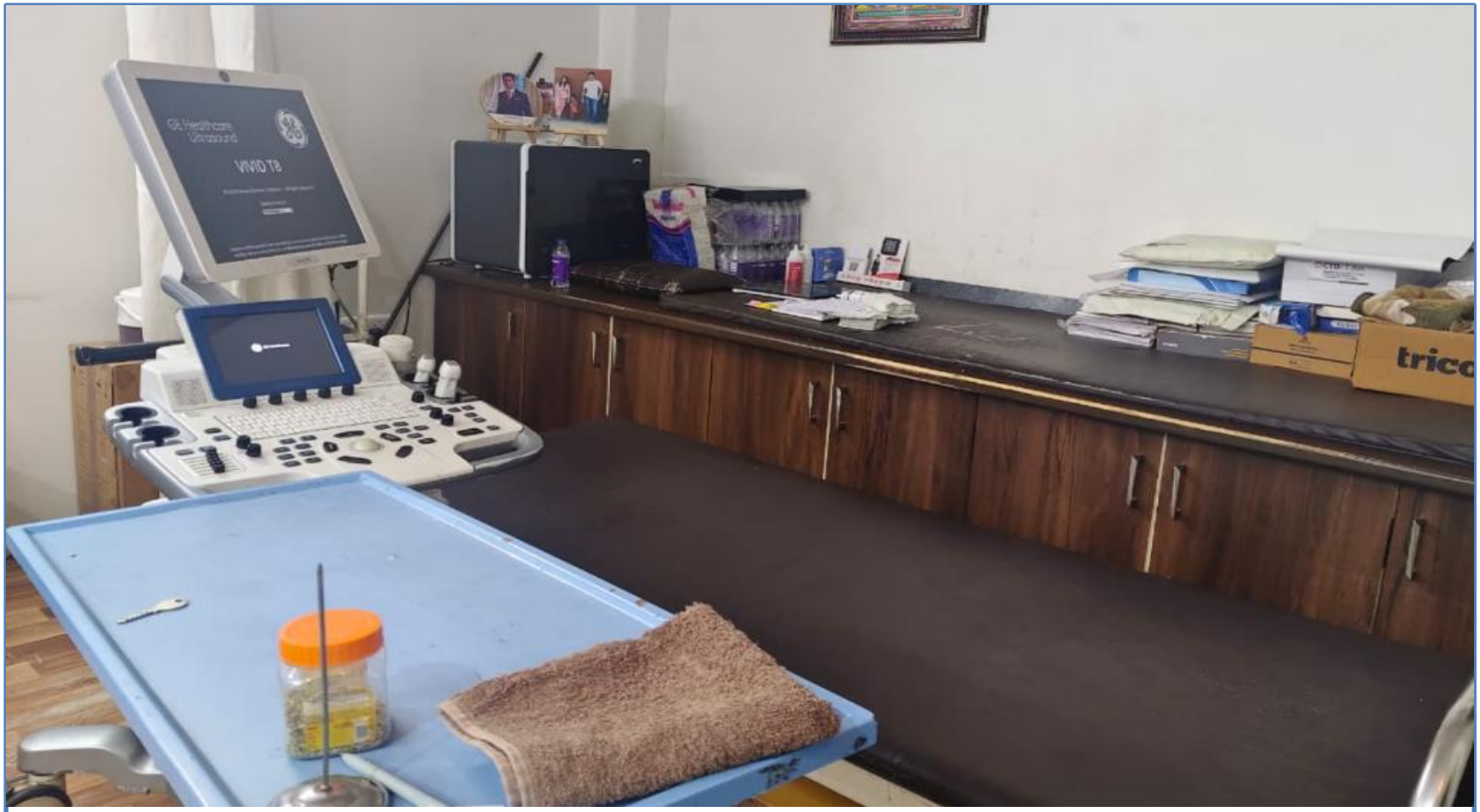
The image shows a manual patient admission form currently used at Shree Omkar Clinic. This paper-based form captures comprehensive patient information during the admission process. Additionally, there is a separate section for expired patients to document death details, autopsy, and post-op information. This manual form highlights the need for a digital system to reduce paperwork, prevent data loss, and make patient records easily searchable and accessible.

2. Literature Review



The Image shows the admit patient interface where receptionist can record and manage admit patient details

2. Literature Review



The Image displays a 2D Echo machine used at Omkar Clinic for cardiac diagnostic purposes.

3. Scope of the project

A. In-Scope :-

- **User Roles and Modules:**

Receptionist :- Register new patients.

- View total no. of Patients and Apponitments.
- Search Patients by Patient's details.
- Admit registered patients.
- Assign laboratory tests and view all bookings.

Doctor :- View no. of pending appointments and daily appointments.

- Can set their daily Schedule and OPD Schedule.
- View their recent activities.
- Can manages their Profile.

Admin :- Manages Clinic Staff and Doctor.

- View and Manages no. of Beds.
- Stores all Registered Patients.

3. Scope of the project

- **Functional Scope:**

- Patient Registration and Maintenance of Patient Records.
- Appointment Scheduling and Management of Doctor.
- Patient Admitted and Discharge management.
- Laboratory test assigning and tracking
- Clinic Staff Management.

B. Out-Scope:-

- No online appointment booking by patients themselves.
- No integrated payment gateway or invoice generation.
- Test booking is available, but no lab report generation or integration.
- No medicine inventory or prescription dispensing.

4. Methodology

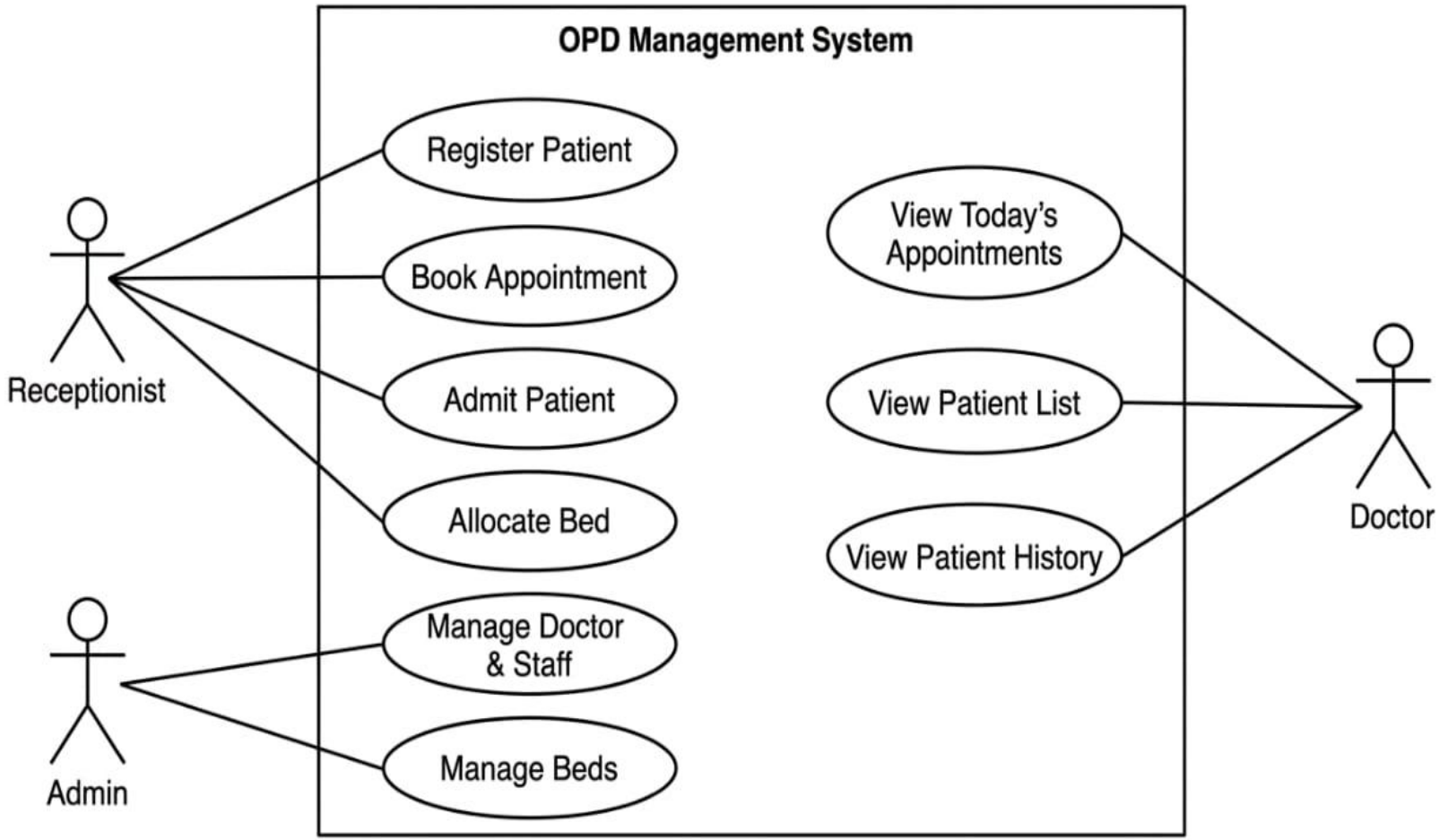


Fig 4.1 Use-Case Diagram for OPD Management System

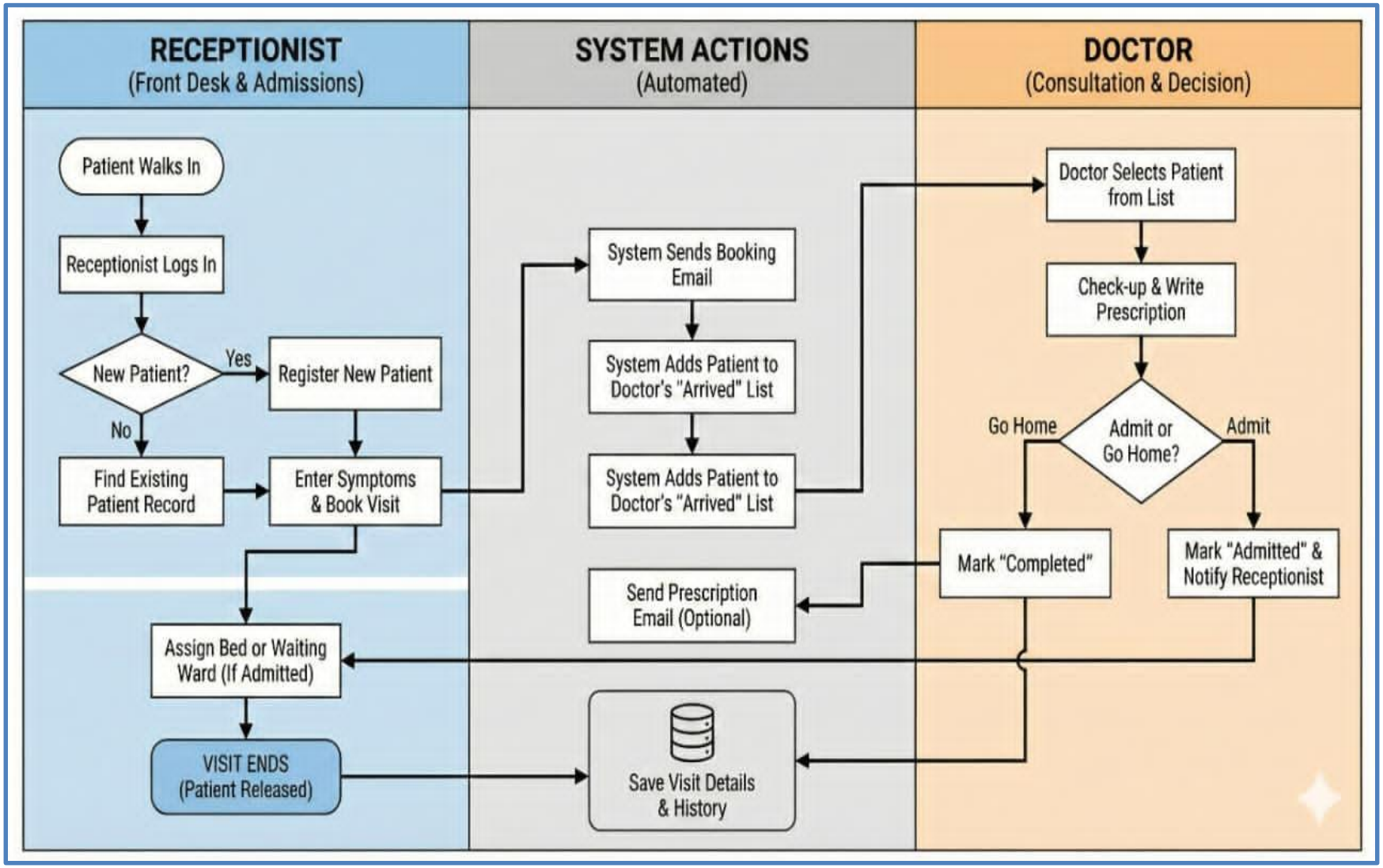
4. Methodology

- **Development Phases:-**
- **Phase 1 –** Requirements: Stakeholder interviews, use case modeling, requirement analysis
- **Phase 2 –** Design: Database schema design, UI/UX prototyping, role-based architecture planning
- **Phase 3 –** Implementation: Frontend coding (React), Backend API development (Node.js), JWT authentication, Email service integration
- **Phase 4 –** Testing: Unit testing (Jest), Integration testing (Postman), User Acceptance Testing with clinic staff
- **Phase 5 –** Deployment: Local server deployment at Omkar Clinic, staff training sessions

➢ **Technology Stack:** MERN (MongoDB, Express.js, React, Node.js) with HTML, CSS, and JavaScript.

➢ **Descriptive Analytics:** Patient volume, appointment status, peak hours analysis

5. Details of designs, working & processes



5. Details of designs, working & processes

A. Receptionist Working:- The Receptionist module serves as the entry point for all patients visiting Omkar Clinic. The receptionist checks whether the patient is new or existing. For a new patient, the system displays a registration form where and for an existing patient, the receptionist can book a new appointment and if required Admit the patient.

B. Doctor Working:- The Doctor module is designed specifically for Dr. Pranjal Patil, the cardiologist at Omkar Clinic, to manage patient consultations seamlessly. From here, Doctor can view the list of patients who have arrived and are waiting for consultation, along with the total count of pending and completed appointments.

C. Admin Working:- The Admin module provides complete supervisory control over the entire OPD management system. The module also enables resource management, allowing the admin to view real-time availability of clinic resources including beds and Additionally, the admin can monitor overall system activity.

6. Results and Applications

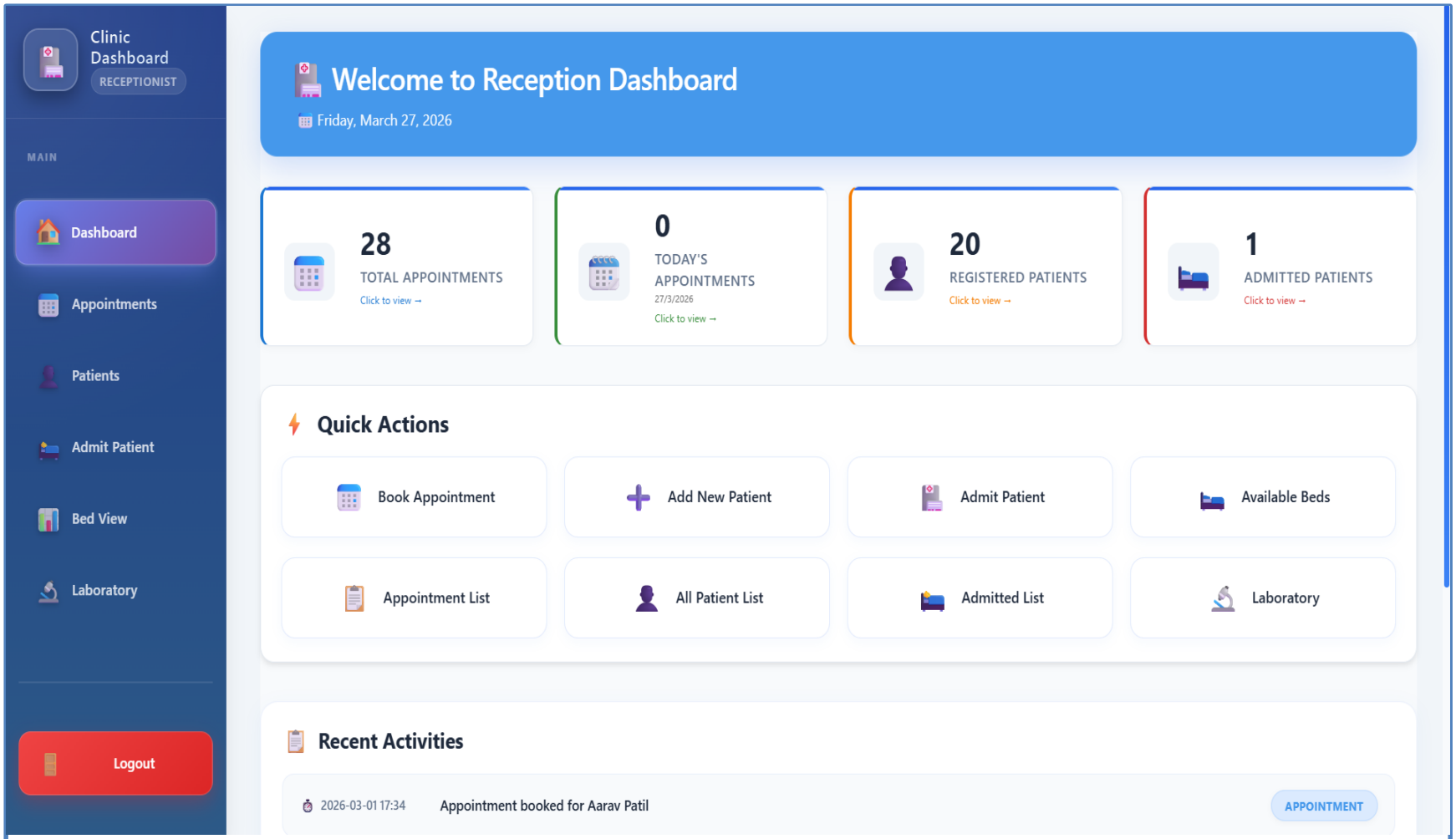


Fig 6.1 Dashboard page for Reception Module

6. Results and Applications

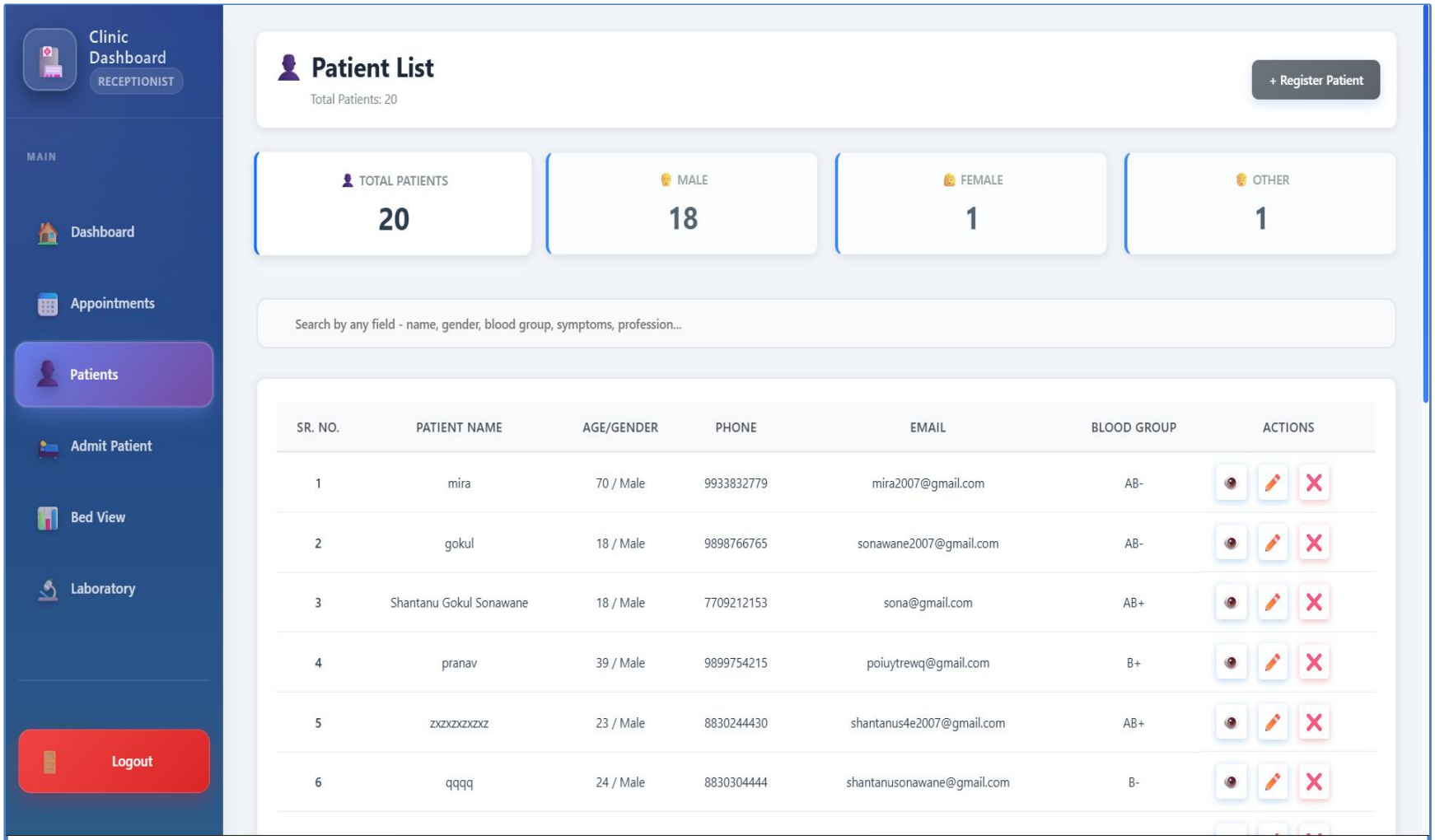


Fig 6.2 Patients List page for Reception Module

6. Results and Applications

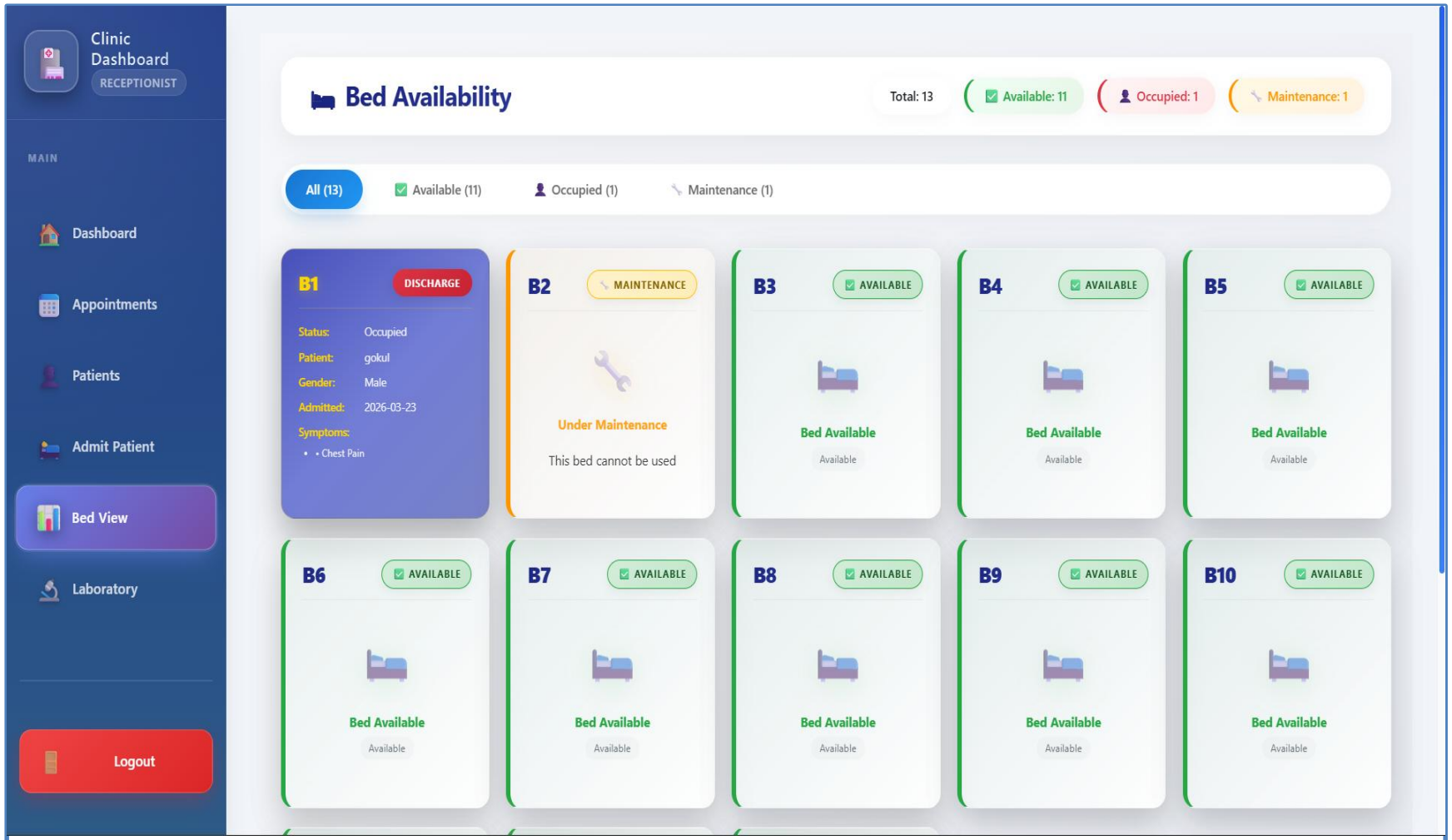


Fig 6.3 Bed View for Reception Module

6. Results and Applications

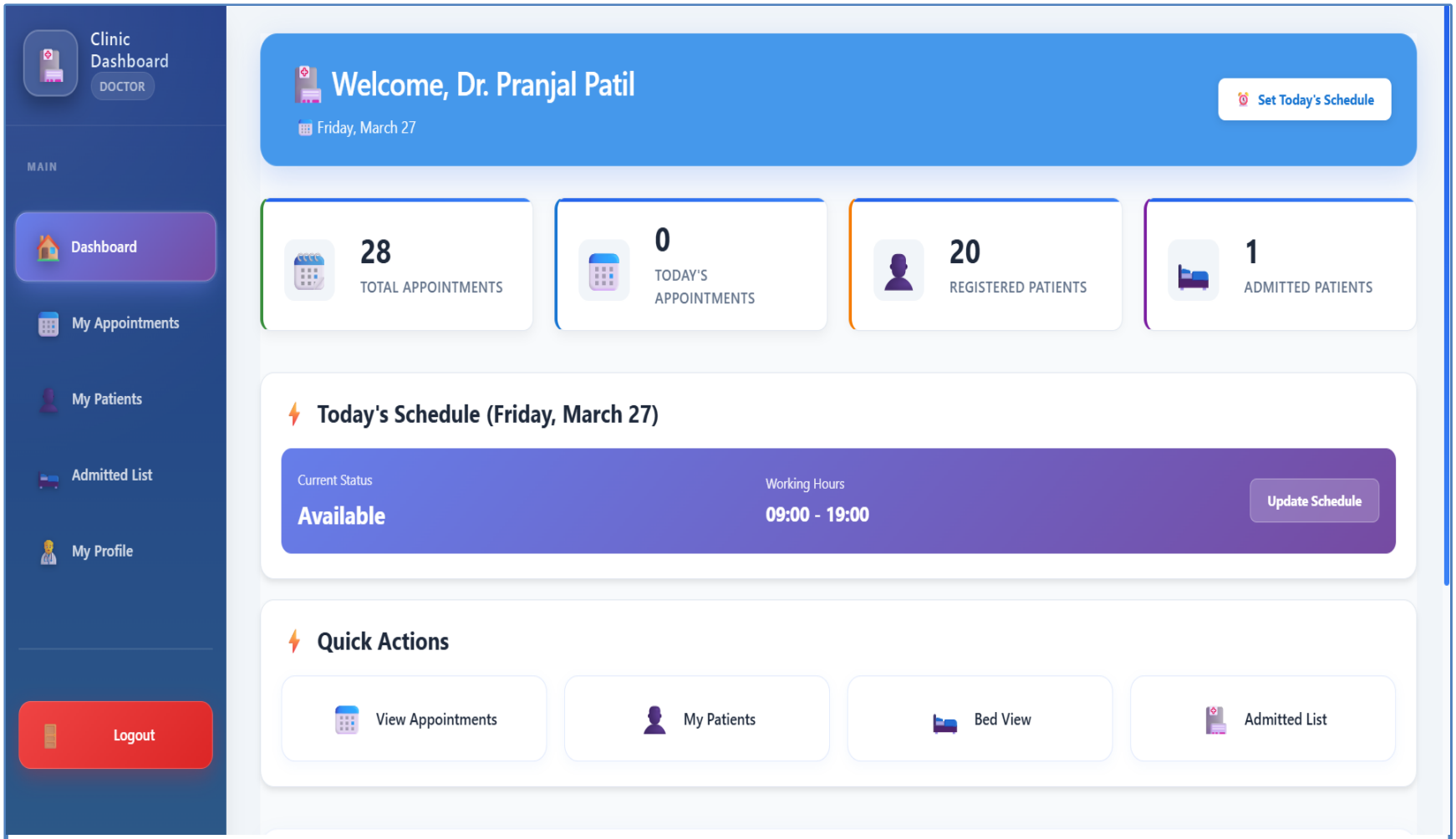


Fig 6.4 Dashboard page for Doctor Module

6. Results and Applications

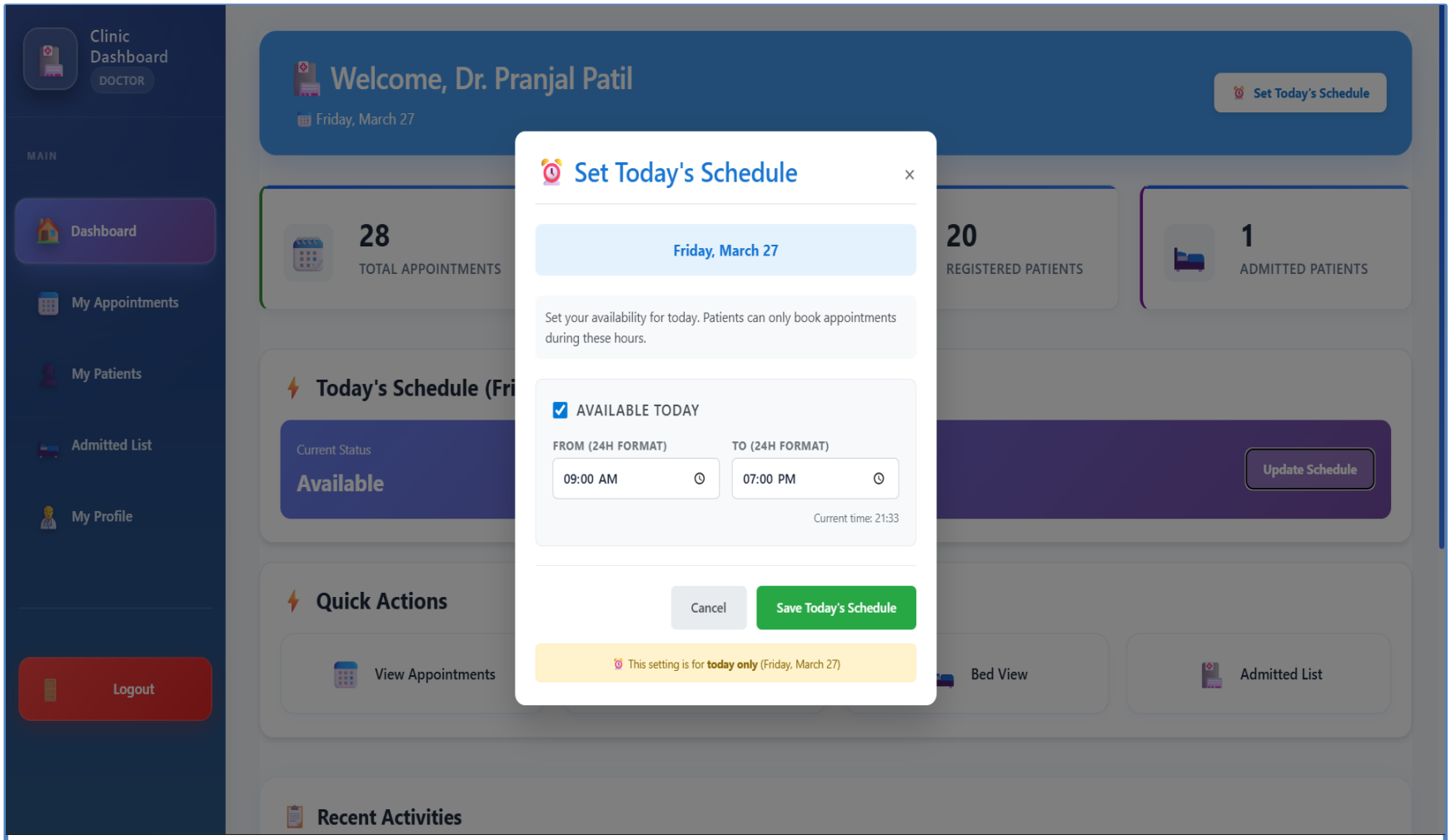


Fig 6.5 Time Scheduling for Doctor

6. Results and Applications

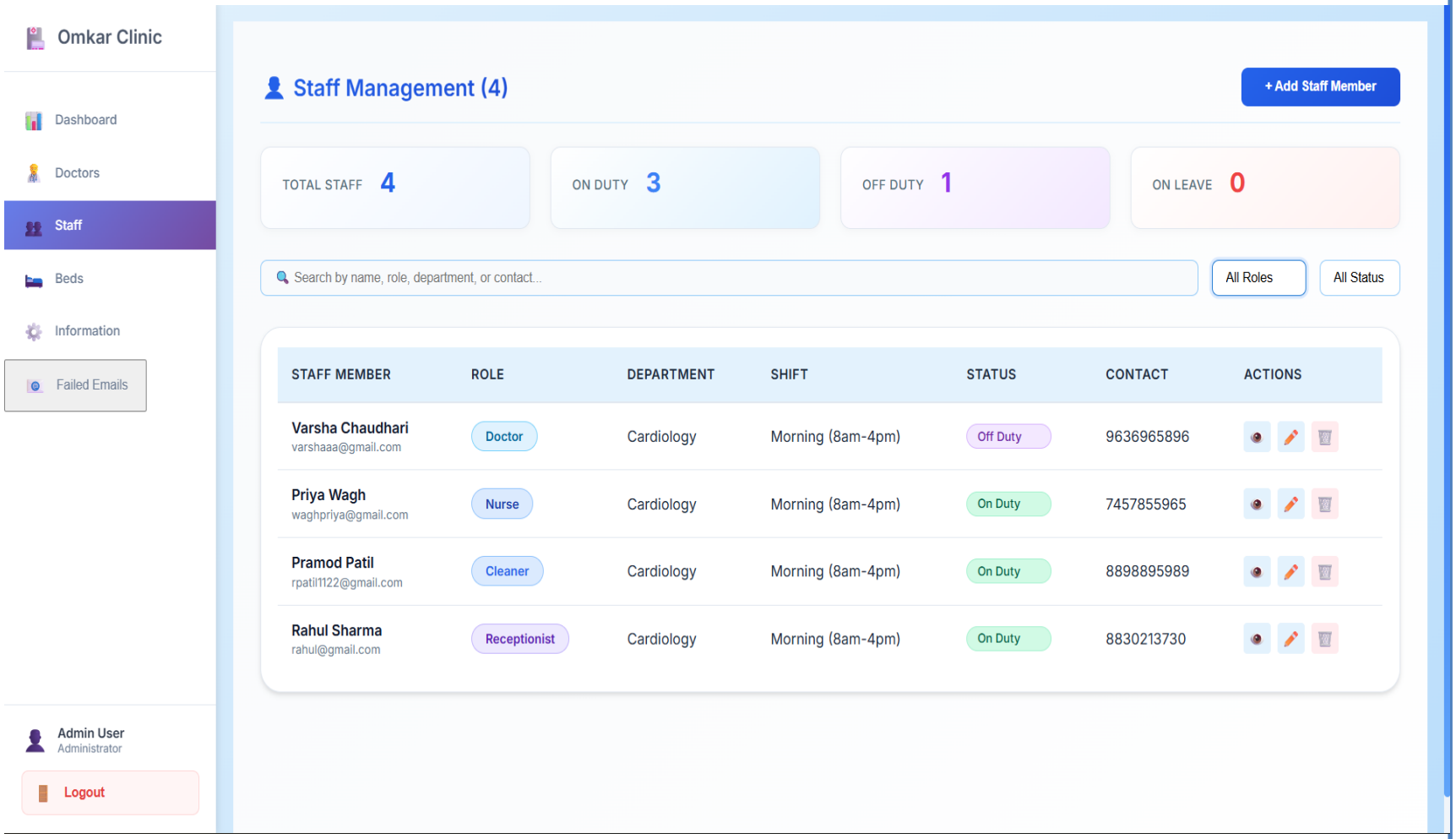


Fig 6.6 Staff Management page for Admin Module

6. Results and Applications

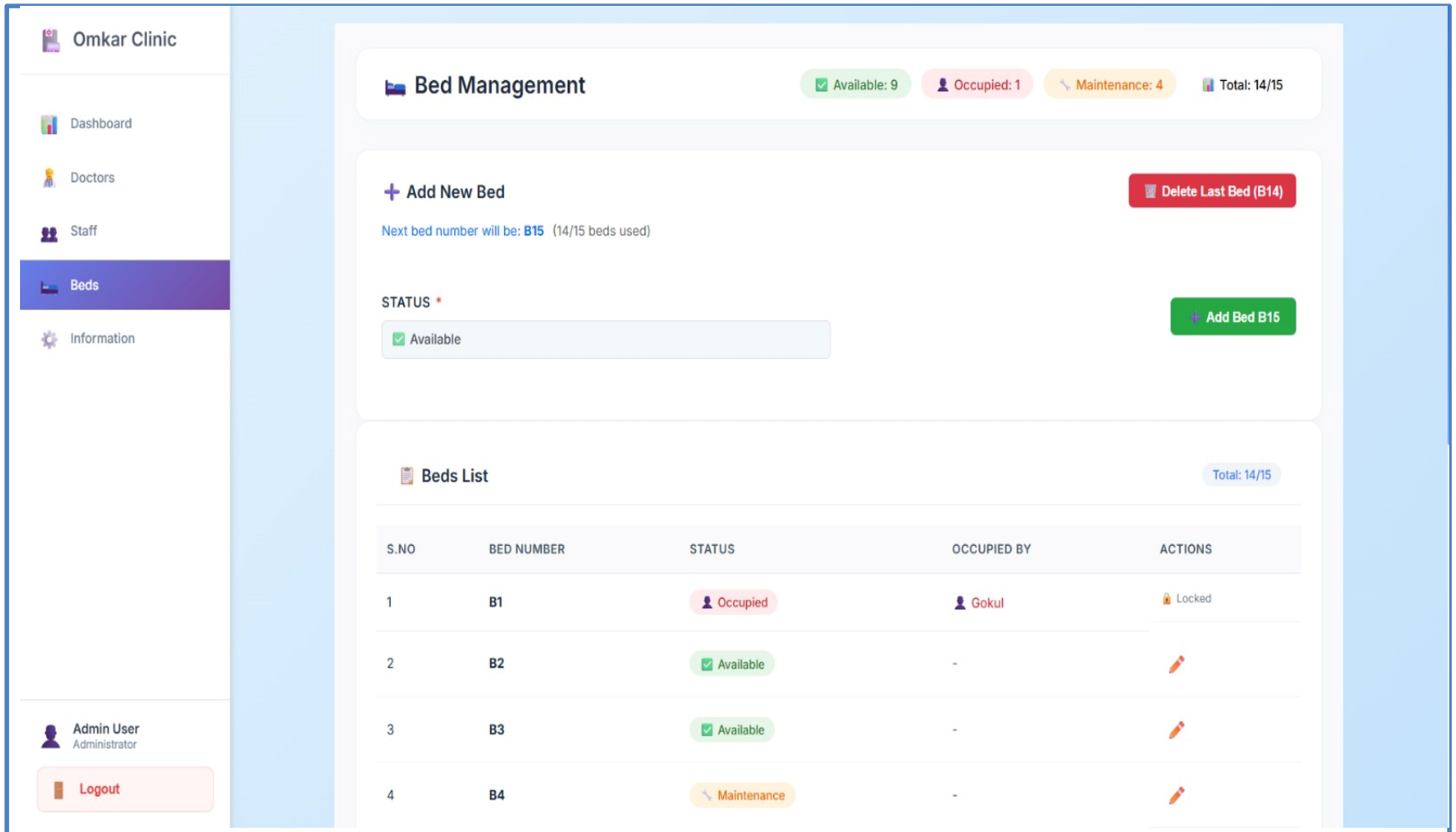


Fig 6.7 Beds Management page for Admin Module

6. Results and Applications

Omkar Clinic <medicarehospital800@gmail.com>
to me

Subject: Welcome to Omkar Clinic - Appointment Confirmed

Dear Shantanu Gokul Sonawane,

Thank you for registering with Omkar Clinic. Your patient appointment has been successfully completed.

📋 Appointment Details:

- Appointment Date: 27/03/2026
- Appointment Time: 13:45
- Name: Shantanu Gokul Sonawane
- Age: 18
- Gender: Male
- Phone: 7709212153

📍 Clinic Address:
Omkar Clinic, Wadibhokar Road,
Deopur, Dhule

👨‍⚕️ Doctor: Dr. Pranjal K. Patil

☎️ Contacts : 025-62221234

Your medical records have been created in our system. Please bring your ID proof for your first visit.

For any queries or to book appointments, please contact our reception desk.

Thank you for choosing Omkar Clinic!

Best regards,
Omkar Clinic

Fig 6.8 Appointment Confirmation Mail for Patients

7. Conclusion and future scope

- **Conclusion:-**

In Conclusion, The Outpatient Department (OPD) Administration Management System has successfully transformed the manual workflow at Omkar Clinic into a fully digital, efficient, and organized process. By addressing the specific needs of Receptionist, Dr. Pranjal Patil, and Admin, the system reduces patient waiting time, eliminates paperwork, ensures data accuracy, and provides a seamless experience for both staff and patients. This project demonstrates how modern web technologies like React, Node.js, and MongoDB can be effectively applied to solve real-world healthcare administration challenges. The system is fully functional, user-friendly, and ready for daily use at Omkar Clinic for Receptionist, Doctor and Admin.

This system is not just a project completion - it is the beginning of a digital transformation journey for Omkar Clinic with a clear roadmap for future enhancements.

7. Conclusion and future scope

- **Future Scope:-**

1. A secure web portal where patients can log in to view their medical history, past prescriptions, and visit details at their convenience.
2. Functionality to print prescriptions directly from the system for patients who prefer or require paper copies of their prescriptions.
3. Integration with popular payment systems (credit card, debit card, UPI) to allow patients to pay consultation fees online.
4. Functionality for lab technicians to upload digital lab reports directly to patient records for easy access by the doctor.
5. Extending the system to support multiple doctors if Omkar Clinic expands its team in the future.
6. Dedicated Android and iOS mobile apps for patients to book appointments, view history, read prescriptions and receive notifications on their smartphones.

8. References and Bibliography

- **References:**

1. https://healthray.com/
2. https://www.doctorsapp.in/
3. https://www.easyclinic.io/
4. https://github.com/

- **Bibliography:-**

1. Omkar Superspecialist Hospital. Dr. Pranjal Patil, Cardiologist, Dhule.
2. Health Technologies Private Limited. (2016). Care Turtle: Book Online Token & Avoid Waiting at Clinic & Hospitals Retrieved from https://play.google.com/store/apps/details?id=com.careturtle.patient1
3. Tracxn. (2024). CareTurtle Company Profile. Retrieved from https://tracxn.com/d/companies/careturtle/